# Phase 1 Review: Lightweight Spike-Based DOA Sector Inference for Embedded Edge AI

Hossein Molhem

Kennesaw State University
EE 6900 Special Topics: Embedded AI for Autonomous Robots

Research Review Presentation
July 12, 2026

# Presentation Roadmap

1. Motivation and problem definition
2. Phase 1 signal-processing and inference pipeline
3. Dataset and experimental design
4. Baseline and LIF-inspired results
5. Accuracy–latency interpretation
6. Limitations and future work

## Target duration

This slide deck is designed for a 15–17 minute presentation.

# Motivation

- Embedded AI systems operate under compute, memory, energy, and latency constraints.
- Many edge-sensing problems have useful physics-based structure.
- Spike-based inference can represent information as sparse temporal events.
- The project asks whether array-processing features can support spike-based temporal inference.

# Problem Statement

## Phase 1 Question

Can compact beamspace features extracted from a physics-based ULA signal model be converted into sparse spike trains and classified with acceptable accuracy and controllable temporal latency?

- Task: four-class DOA sector classification.
- Scope: one narrowband source, controlled SNR grid.
- Purpose: establish a reproducible embedded-AI baseline, not a new high-resolution DOA estimator.

# Why DOA Sector Inference?

- DOA processing connects electromagnetics, radar, MIMO sensing, and array signal processing.
- Sector classification is smaller and more controlled than continuous-angle estimation.
- Beamspace features provide compact angular signatures.
- The covariance stage creates a natural bridge toward FPGA/HLS acceleration.

# End-to-End Phase 1 Pipeline

ULA / DOA signal model

⇓

Array snapshots with controlled SNR

⇓

Sample covariance matrix $R_{xx}$

⇓

Beamspace energy features

⇓

Nearest-centroid baseline

⇓

Rate-based spike encoding and LIF-inspired inference

# Theoretical Foundation

$$a(\theta) = [1, e^{-jkd\sin\theta}, \ldots, e^{-j(N-1)kd\sin\theta}]^T$$

$$R_{xx} = \frac{1}{M} X X^H$$

$$p(\phi) = b^H(\phi) R_{xx} b(\phi)$$

$$\hat{c} = \arg\min_c \|x - \mu_c\|_2^2$$

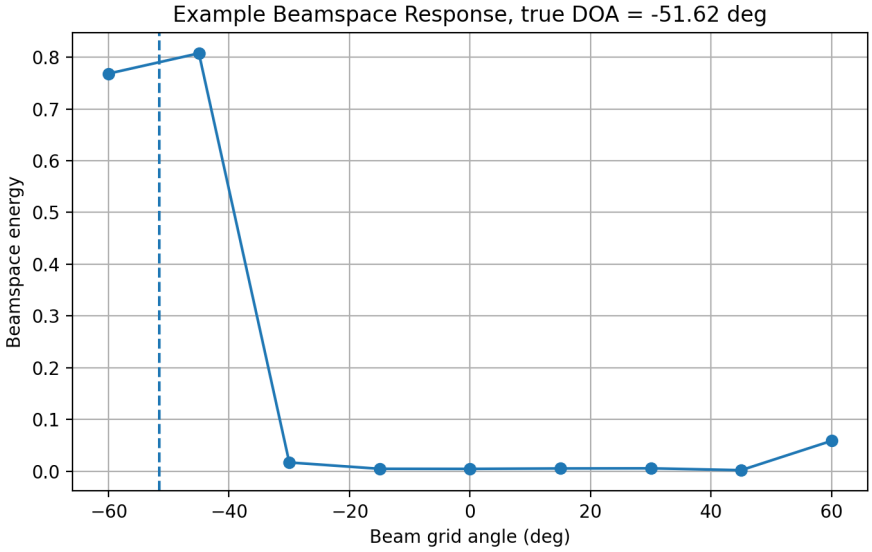
$$v_c[t] = \alpha v_c[t-1] + i_c[t]$$

- Covariance captures spatial correlation.
- Beamspace energy compresses spatial information.
- LIF accumulation creates a temporal decision process.

# Dataset Configuration

| Parameter | Value |
|---|---|
| Array type | Uniform Linear Array |
| Array elements | 8 |
| Snapshots per sample | 128 |
| Element spacing | $\lambda/2$ |
| Source count | 1 |
| DOA range | $-60^\circ$ to $+60^\circ$ |
| DOA sectors | 4 |
| SNR values | $-10, -5, 0, 5, 10, 15, 20$ dB |
| Total samples | 2800 |
| Test samples | 840 |

# Beamspace Feature Example



# Baseline Classifier

## Nearest-centroid decision rule

$$\hat{c} = \arg\min_{c} \|x - \mu_c\|_2^2$$

- Uses continuous normalized beamspace features.
- Deterministic, simple, and interpretable.
- Establishes the non-spiking reference accuracy.
- Overall test accuracy: 0.869048 or 86.90%.

# Spike Encoding and LIF-Inspired Inference

**Rate-based spike encoding**

- Normalized feature values become spike probabilities.
- Larger features generate more spikes over time.
- Produces sparse temporal representations.

**LIF-inspired classifier**

- Class-level membrane evidence.
- Leak controls evidence retention.
- Stable-decision latency becomes measurable.

# Fixed LIF Configuration

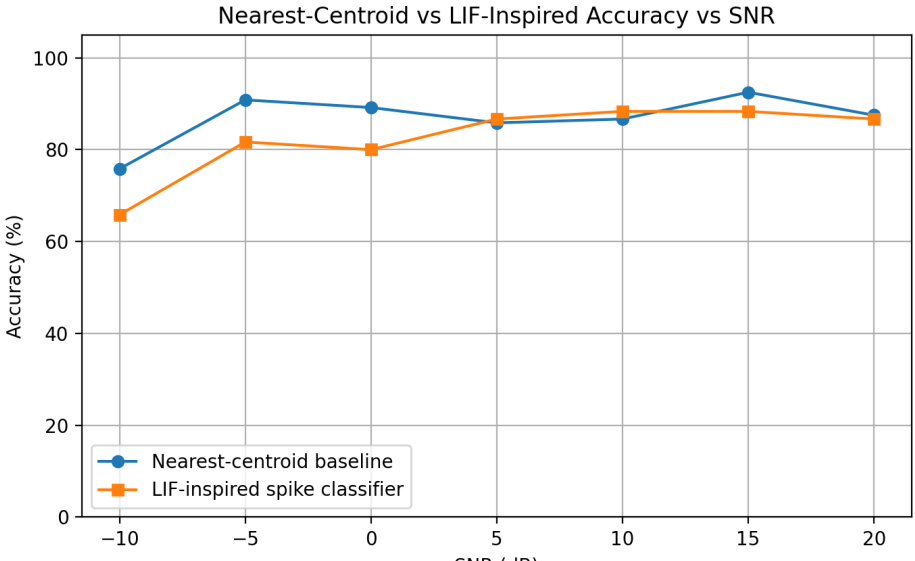
$$v_c[t] = \alpha v_c[t-1] + i_c[t]$$

## Run 003

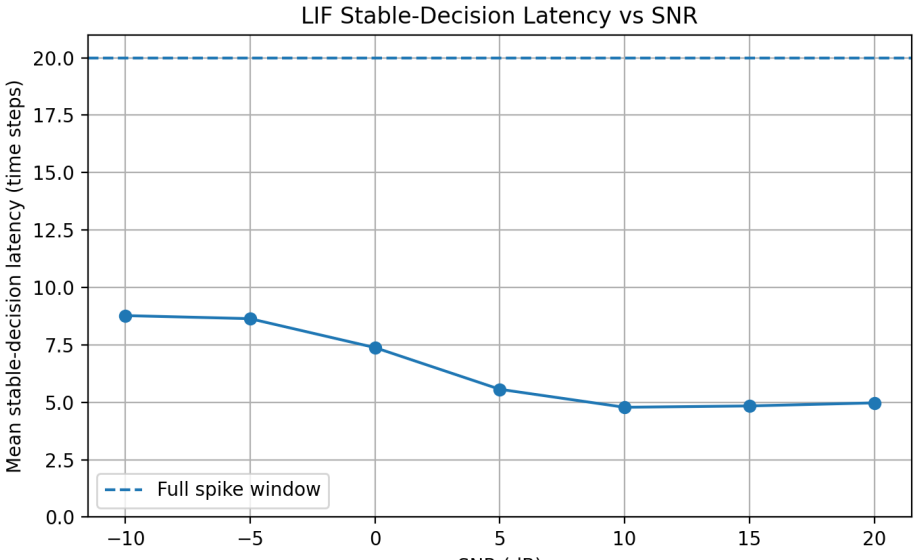
$T = 20$, leak $= 0.97$
Accuracy $= 0.825000$
Mean stable-decision latency $= 6.423810$ time steps

- The fixed LIF path preserves useful classification behavior.
- It remains below the continuous baseline in raw accuracy.

# Accuracy Comparison Versus SNR



# Stable-Decision Latency Versus SNR

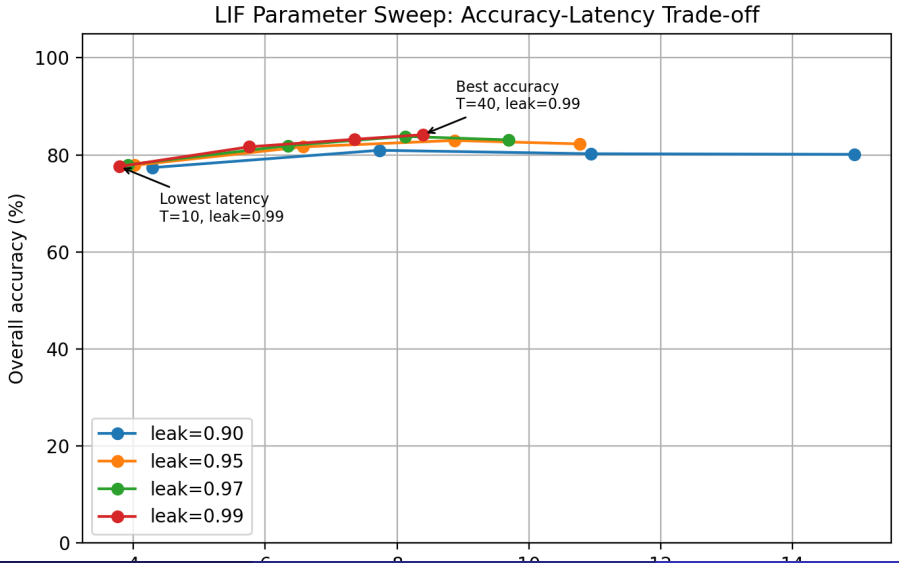


# Run 005 Parameter Sweep

| Case | $T$ | Leak | Accuracy | Latency |
|---|---|---|---|---|
| Best LIF accuracy | 40 | 0.99 | 0.841667 | 8.396429 |
| Lowest latency | 10 | 0.99 | 0.776190 | 3.782143 |
| Fixed Run 003 | 20 | 0.97 | 0.825000 | 6.423810 |
| Baseline | – | – | 0.869048 | – |

The sweep exposes a tunable accuracy–latency design space.

# Accuracy–Latency Trade-Off



# Main Engineering Interpretation

- The spike-based path does not improve raw accuracy over the continuous nearest-centroid baseline.
- This is an expected and useful result because rate-based spikes introduce stochastic quantization.
- The value of the spike path is architectural:
  - sparse temporal inference,
  - measurable stable-decision latency,
  - tunable accuracy–latency behavior.

# Connection to Prior FPGA/HLS SCM Work

- Prior work implemented a Vitis HLS sample-covariance-matrix accelerator for DOA preprocessing.
- That project targets the covariance stage:

$$R_{xx} = \frac{1}{M} X X^H$$

- The current Embedded AI project builds on the same covariance-based DOA foundation.
- Together, the two repositories create a path from DOA preprocessing to hardware-aware spike-based inference.

# Limitations

- Single-source narrowband ULA model.
- Sector classification rather than continuous high-resolution DOA estimation.
- Rate-based spike encoding introduces stochastic variation.
- LIF classifier is a simplified class-level temporal accumulator.
- Current implementation is Python-based and not yet fixed-point or deployed on FPGA.

# Conclusion and Future Work

## Conclusion

Phase 1 establishes a reproducible DOA-sector embedded-AI baseline. The continuous beamspace classifier provides strong accuracy, while the spike-based path provides sparse temporal inference and explicit latency control.

- Baseline accuracy: 86.90%.
- Best LIF accuracy: 84.17%.
- Next steps: richer DOA cases, fixed-point analysis, PYNQ/Vitis HLS refactoring, and hardware-aware deployment planning.

# Closing Message

Phase 1 converts a physics-based DOA feature pipeline into a latency-aware embedded-AI inference baseline.

The main contribution is not higher raw accuracy; it is a controlled bridge toward sparse temporal and hardware-aware edge inference.